Salamander Habitat Suitability Study

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Acknowledgements

This report has been so fun to put together, drawing on our different areas of expertise and background. This we hope is a comprehensive understanding and plan for the new ventures being pursued at the Laurie Lawson Education Centre Property, owned and managed by the Northumberland Land Trust.

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This has been a labour of love and we hope to work with you in the future!

Sarah Bencic and Shantanu Dutt

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Introduction

This project was developed for the Northumberland Land Trust (NLT) to assess and determine the suitability of the Laurie Lawson Education Centre property in Cobourg; for possible salamander introduction. In the initial research for this project many experts and expert literature sources were consulted to develop methods of assessment and to layout our onsite research. From this research we established that there are many factors which might affect survival of salamanders in an established population, and that would affect the suitability of habitat for translocation of salamanders. The substantial number of vernal pools on the property was the main reason for the Northumberland Land Trust's request for research and an assessment into the property's suitability.

Given this unique property feature our research focused on three main species of salamander; Spotted (*Ambystoma maculatum*), Blue Spotted (*Ambystoma laterale*) who utilize these specific habitats for breeding and larval development (Karraker & Gibbs, 2009), and the third species considered would be the Red Backed *Plethodon cinereu). Plethodon cinereu* would be an excellent candidate for relocation because of its generalist abilities (Maerz et al., 2005). In addition, we also considered a newt that is and was known to occur in the same search radius: The Central Newt (*Notophthalmus viridescens louisianensis*). The *Notophthalmus viridescens*, though not a salamander, was also evaluated for potential for relocation to the site because of it's generalist abilities like the *Plethodon cinereu*. Jefferson (*Ambystoma jeffersonianum*) salamanders could also benefit from this type of habitat and are endangered. If the habitat suits, then this would likely result in some complications in permission to relocate these organisms.

The research for this project was broad in nature given the research objectives; understanding habitat requirements of the salamanders and to develop methods for habitat assessment and monitoring for the NLT. The primary focus was to understand what would make a suitable habitat for the species of salamanders to use if they were to be relocated. The primary suitability qualities measured were: soil quality, pH, water quality, canopy cover, habitat connectivity, vernal pool abundance, and longevity of the vernal pools. These qualities were targeted as necessary in the literature and during the expert interviews conducted prior to taking measurements on the property.

The people consulted were: Madison Wikston - Masters Student at Trent, Patrick Moldowen – PhD candidate at University of Toronto, Mike Oldham– Natural Heritage and Information Centre and Tina Frigden at Trent University. All four of the consulted experts emphasized the importance of connectivity above all other qualities as most important for survival of salamanders in an area. The vernal pools on the Northumberland property are numerous and though would likely support development of the larval salamanders, there needs to be sufficient upland habitat connectivity to support the adult salamanders. Additional upland habitat might be provided by creating corridors to the nearest undisturbed forests habitat This could be done by connecting farmers with government subsidies for creating buffers between the surrounding agricultural land and the NLT property. Implementing a protected riparian zone along Cobourg Creek, which runs through the property and using it as a habitat corridor would be the simplest method of acquiring connectivity.

After the initial survey of the property it was quite surprising to note that no salamanders were found. The overall composition of the area, the forest canopy and topography are all categorically what is preferred by salamanders. More suitability measurements were thought to be needed to better understand why there are no salamanders

currently occupying the space. The property is reclaimed agriculture land and the former agricultural activity and extreme fragmentation may have eliminated salamanders. Because of this they have likely not reoccupied this space. The wood frogs observed on the property are more adapted to travel longer distances and not dry out. From our expert interviews and initial research, it is expected that if more continuity could be created between upland habitat and this property the salamanders may reoccupy the area on their own. The nearest possible space where other salamanders might be found is separated by large areas of farm land, roughly 4km away, East of the Hamilton Administration Office. This is based entirely on the presence of green habitat and water bodies. The Ontario Reptile and Amphibian Atlas does not, however, show any confirmation of presence of relevant species. Hence, while the property has potential to support salamanders, a survey for potential donor salamander populations will be required. The Laurie Lawson property could be potentially suitable for introduction of salamanders if there was greater connectivity to necessary land habitat, but currently travel distance is a limiting factor for salamanders.

Potential Salamander Species

Using the Ontario Reptile and Amphibian Atlas Program we used a search radius of 50km from the site on the Canadian side of Lake Ontario, due to it being a natural barrier, to find salamander species that are currently or have historically been present in the region. The following three salamander species have been recorded to be in this search perimeter: *Plethodon cinereus*, *Ambystoma laterale*, and the *Ambystoma maculatum*. There is also the newt sub-species; *Notophthalmus viridescens louisianensis*. All four have been designated as Specially Protected Amphibian under the Ontario Fish and Wildlife Conservation Act.

Blue- spotted Salamander ~ *Ambystoma laterale*

The *Ambystoma laterale* in North America have been shown to migrate up to 219m from breeding vernal pools, averaging around 113m. Individuals use small mammal burrows as terrestrial refuge during the day and the availability of colder areas for shelter is possibly a limiting feature for them on the landscape. The presence of the species is correlated positively with sapling and log cover with their populations. *Ambystoma laterale* preferences range from deciduous stands and leaf litter to conifers. Soil moisture and leaf litter are primary indictators of shelter in the landscape (Faccio, 2003: Ryan & Calhoun, 2014). Egg masses are attached to woody vegetation surrounding pools. Egg masses are also seen to be more abundant in pools that have hydroperiods of 8 to 12 months, mostly for 10 to 11 months. There is a significant negative effect noted in the literature where roads are too closely situated to the pools used by this species (Egan & Paton, 2004).

Eastern Red-backed Salamander ~ *Plethodon cinereus*

The Eastern Redbacked Salamander is a terrestrial amphibian and does not have an aquatic phase like many other salamanders. This species of terrestrial salamander is known to have a positive correlation with quantity and quality of woody debris, and canopy cover (Brooks, 1999). While the salamanders are known to be territorial, sometimes individuals might float between territories. This can result in an overlap of intersex communities (Mathis, 1991). There have been no observed significant immediate changes to populations from reduction of canopy cover (Messere & Ducey, 1998); though specific silvicultural practices show significant effect on populations (Harpole & Haas, 1999). In lab setting, low pH had been shown to reduce sodium and cause loss of mass in Redbacked Salamanders (Frisbie & Wyman, 1991). Redbacked Salamander populations in Southern Connecticut, U.S.A,

remained relatively unaffected by landscape fragmentation, occupying any available habitat (Gibbs, 1998).

Spotted Salamander ~ *Ambystoma maculatum*

Spotted Salamanders have shown high levels of site fidelity when it comes to breeding vernal pools, then they will disperse up to 500m from them ignoring other pools even if encountered. Displacement, however, causes adults to choose new pools for breeding (Shoop, 1968). The individual eggs have a symbiotic relation with the algae *Oophila amblystomatis* which have been shown to keep the eggs saturated with Oxygen in the presence of sunlight (Bachmann et al., 1986). Incubation temperature affects how soon the larvae emerge. There is a negative correlation of larvae body size and development stage with the incubation temperature. Incubation up to roughly 10℃ seems to be the upper threshold before larvae size and development shows significantly negative influence (Voss, 1993). The eggs can take up to nearly 60 days to hatch with the larvae often overwintering in the pools (Whitford & Vinegar, 1966).

Embryonic survival in pools has ben shown to be negatively correlated with Aluminum concentrations. On the other hand, embryonic survival was positively correlated with magnesium concentrations and water temperature (Albers & Prouty, 1987). Lower levels of pH increase the dissolved organic content and biologically available aluminum. pH<4.8 and inorganic monomeric Al at concentration of 37x10-6g/L becomes toxic, making the eggs more susceptible (Clark & Roland, 1985). Experiments show larval mortality before metamorphosis at pH 4.2 and lower (Sadinski & Dunson, 1992). Larvae have shown mortality at concentrations of dissolved oxygen of 1.5 ppm but no mortality at 4.0 ppm of dissolved oxygen. They have adaptability to low dissolved oxygen levels and their metabolic rates are positively correlated with temperature (Branch & Taylor, 1977).

Caddisfly larvae of *Banksiola dossuaria* have been shown to prey on egg masses of spotted salamanders (Stout, Stout & Stihler, 1992). Egg masses are also prey to wood frog, *Rana sylvatica*; though whiter egg masses show lower predation rates (Petranka, Rushlow & Hopey, 1998). Adults of this species have critical thresholds with direct forest cover: depending on the habitat size; ranging from 30% to approximately 50% of forest cover, and from a 100m to1000m habitat size, respectively. Because of this, salamander populations seem to show a negative effect (population decrease) when the forest cover is less than the range (Homan et al., 2004).

Eastern Newt (*Notophthalmus viridescens*)

Eastern newts consist of four sub-species and are also called 'red-spotted newts', though it is also the name of one of the four sub-species. The Central Newt (*Notophthalmus viridescens louisianensis*), is the primary sub-species in the region but literature exists primarily for the red-spotted sub-specie *(Notophthalmus viridescens viridescens)*. The species uses vernal pools for breeding, akin to the salamanders. Newts tend to metamorphose into terrestrial 'efts' and then finally, in 2 or 3 years, into aquatic adults but it is not a mandatory phase in their life. Observing a population of the Red-spotted sub-species in Nova Scotia, the adults show site fidelity when it comes to breeding. Hence, the populations showed two migrations. An early spring migration for breeding and an August-September migration for when, both efts and adults, move into their terrestrial hibernacula though some adults may remain aquatic during the winter. While the adults show site fidelity, the juveniles migrate to new pools, making them a colonizing species. The success from the pools showed the strongest effect on the outcome of the overall population. (Gill DE, 1978)

The adults tend to colonize slow moving streams as well while the eft will prefer forest floor litter and logs for shelter. They have a lifetime of about 10 years, and are poisonous to many predators except Garter snakes. They eat invertebrates, egg masses and even other juvenile salamanders. However they are just as sensitive the preferred salamander species, to water chemistry as well (Ontario Nature, *[date unknown]*)(Roe A & Grayson K, 2008).

Suitability Measurements

Soil Quality and pH

Though developing larval salamanders are completely aquatic until emergence, the pool quality is directly influenced by the soil and topography surrounding the pools. The soil composition on the property is on average 12.4% organic matter in the top 5cm of the soil with an average pH of 6.85. Literature suggests that vernal pool dependant salamanders prefer spaces where the soil is loose enough to have small mammals burrowing (Faccio, 2003: Ryan & Calhoun, 2014). The mammal holes afford them some shelter when migrating and searching for food. Rotting logs and leaf litter are also important, and the tree stands covering most of the property is preferred by many of the considered species, though deciduous tree stands are preferred over coniferous tree stands (Faccio, 2003: Ryan & Calhoun, 2014). On our initial survey of the property many small mammal holes were observed and there was a noticeable number of preferred habitat qualities including rotting logs and leaf litter.

Salamanders are somewhat sensitive to the pH of soils, especially in relation to its effects on the water in the pools for growing and developing larvae (Clark & Roland, 1985). Some fluctuation in pH could be expected because of the substantial amounts of rainfall experienced in the spring this year. In other years with more of an average rain fall this would likely not be an issue. The main concern for the NLT pools in relation to pH will be in the water volume and the soil surrounding the pools, as well as, the relationship with dissolved oxygen (Clark & Roland, 1985).

Water quality

While assessing pool longevity and other measurements this summer the quality of water in the many pools was also measured. Salamander larvae are very sensitive to water quality especially with regards to road contaminants (Turtle, 2000). The vernal pools on this property and in general tend to be quite small and are simply not able to reduce and dilute contaminants that end up in the water through the top soil layers, direct runoff and deposition (Turtle, 2000). For this reason, we would recommend if transplanting larval salamanders to not put them in areas close to the property's southern border. The more southern side has a few pools close to the road, but the ones closest will likely be somewhat affected by runoff from the steep slope coming from the road (Figure 1). De-icing chemicals were identified by Turtle's (2000) paper as very damaging contaminants. In Canada though road salts are usual for de-icing and an anti-slip for winter conditions (Environment Canada, 2003). The utilization of a salt and sand mixture is regarded as less damaging than only salt but, it would still need to be accounted for when deciding placement of salamanders (Turtle, 2000). There is a naturalized vegetative barrier between the road and the forested areas and a ditch that would likely help to mitigate some of the interference (Semlitsch, 1998).

The overall water quality is dependant on runoff but also the atmospheric deposition which could be highly determined by developments upwind at the property (Turtle, 2000). Thankfully much of this atmospheric deposition can be intercepted by forest canopy. Buffering zones could thus, be key for management going forward because of the high likeliness of runoff pollutants from both the road and the surrounding agricultural land. Fertilizers used in agriculture in southern Ontario include nitrate-based fertilizers. These are highly toxic to salamanders especially when in their developing larval stage (Marco et al., 2001). Better information on fertilizer impacts is available regarding frogs, especially wood

frogs, which have similar biological and environmental requirements for life as the vernal pool breeding salamanders (Marco et al., 2001).

Canopy Cover

There is a relationship between canopy cover and the longevity of vernal pools because of evaporation. For salamanders, canopy cover is also needed for survival. Adult spotted salamanders have critical thresholds for forest cover, depending on the habitat size as mentioned previously. There is also a positive correlation to sapling and log cover, which is related to number of trees in the stand. Also, as previously mentioned, there is a preference for deciduous stands and for leaf litter instead of just conifer needles, with soil moisture and leaf litter primary dictators of shelter in the landscape (Faccio, 2003: Ryan & Calhoun, 2014). The Laurie Lawson property is a good mix of coniferous and deciduous trees all over the property, some areas more populated with one or the other.

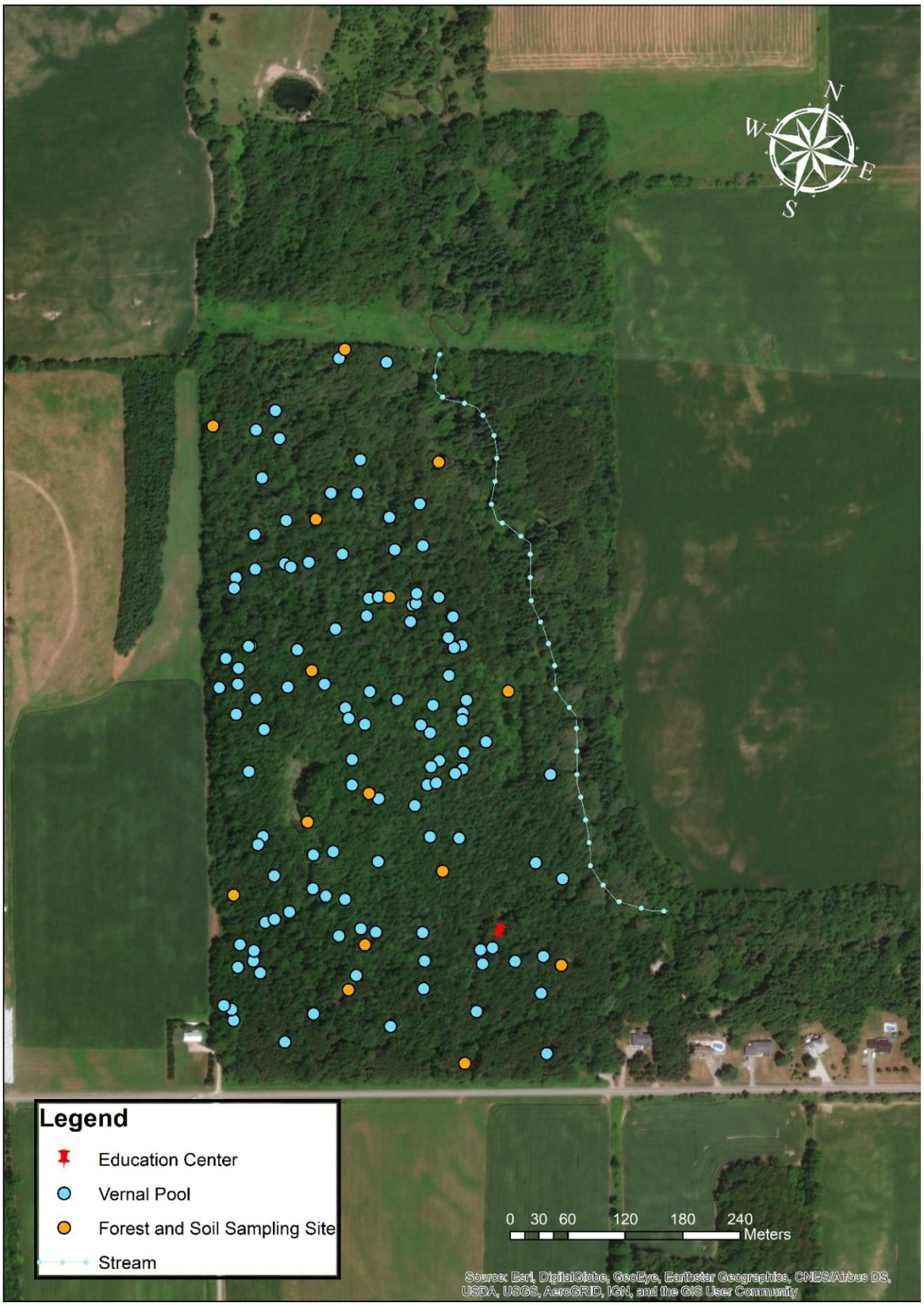

Figure 1. Map of Laurie Lawson Education Centre Property, outlining the surveyed vernal pools and sampling sites for soil and forest canopy cover. Refer to figure 3A in appendix 3, for a more detailed version of the map.

Connectivity of Habitat

This suitability measurement, connectivity of habitat, was one of the most discussed during expert interviews at the initial undertaking of the project and is something that will be most predominant in the recommendations for the final report (Appendix 1). Tina Fridgen, a professor at Trent said, “You’d have to create a pathway for them to migrate on their own.”. Patrick Maldowen, a PhD candidate at the University of Toronto, also specified that, “Connectivity would be the biggest issue other than having basic appropriate conditions”. Mike Oldham at the Natural Heritage and Information Center specified that the Ministry of Natural Resources and Forestry does not deal with moving animals and also mentioned, “You could look at habitat continuity and connectivity to try to help facilitate the salamanders naturally migrating to that property.”. The consensus was, from all the people consulted, that it would be important to promote connectivity and facilitate natural migration.

In the literature reviewed, a study of Spotted Salamander populations (Zamudio, 2007) shows that due to pool fidelity and smaller migration distances, the salamanders can isolate themselves and when this happens a lot of inbreeding occurs around the same pools, resulting in populations being significantly skewed away from random mating. This creates genetically unique populations and genetic diversity across different populations. Populations closer than 4.8km, showed exchange of significant genetic material. Hence, a meta-population of populations with significant population inbreeding while also having some interaction with other populations, creates genetic diversity across the landscape and creates opportunities for rescue and recolonization.

The land owned by the NLT is surrounded by agricultural land, as well as a road to the south and the 401 Highway beyond that (Figure 2). Though the property has many necessary features for a diverse habitat, and enough habitat to sustain many amphibian

species, members of the NLT noted that only within the current year have frogs have made an appearance. There were no signs of salamanders during our visits, though wood frogs were calling in abundance in the spring. Wood frogs require many similar biological and environmental conditions as the vernal pool breeding salamanders and they prey on salamander larvae (Eagon & Paton, 2004). The suitability measurements assessed this season verify that the property is suitable for potential salamander habitat. However, the lack of connectivity requires specific attention.

Adult salamanders using the pools for breeding and then the juveniles for feeding post-metamorphosis will require adequate upland habitat and migration space to fully establish a population (Homan et al., 2003). The forested property allows for a significant amount of leaf litter and rotting logs at varying stages, that could potentially support the salamanders for general foraging and nesting. The connection to other similar naturalized upland habitats would be necessary going forward for both foraging and migration needs, and for encouraging interbreeding and genetic diversity (Homan et al., 2003). The forest cover during migration has been shown to reduce stress and improve overall health of the adult salamanders (Homan et al., 2003). Road crossings can be lethal and exceptionally stressful to the health. This is quite difficult to reduce in Southern Ontario, but where possible effort to increase forest cover would be valuable (Homan et al., 2003). With regards to genetic preservation and health, it would be in the NLT's best interest also to create the possibility of interbreeding through connective corridors (Gibbs & Reed, 2008). One risk with relocation of organisms would be a migratory bottleneck, which could ultimately harm the population and others with which they might interbreed (Gibbs & Reed, 2008).

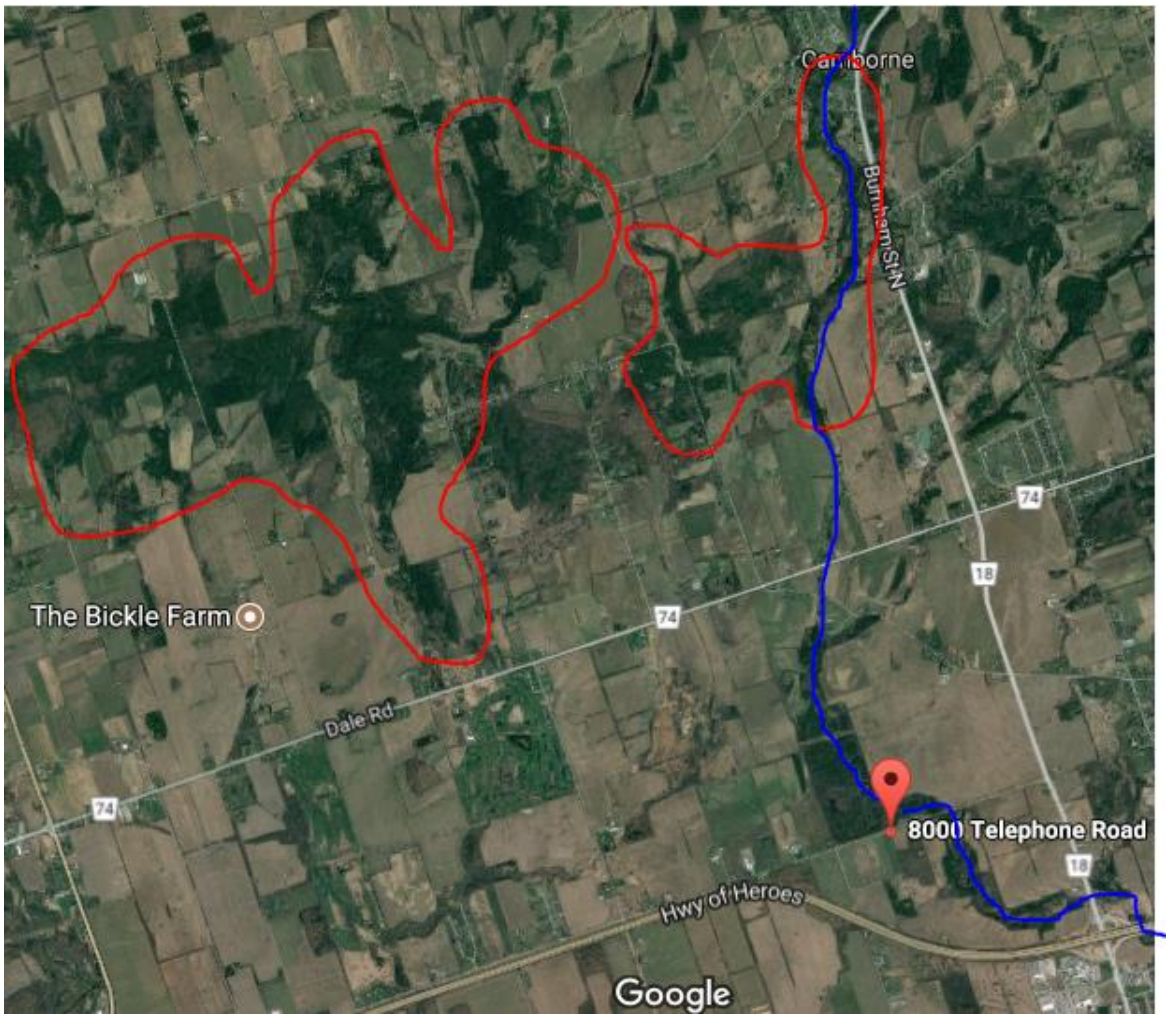

Figure 2. A map of surrounding areas that could potentially be connected with the current property (outlined in red), and the Cobourg Creek that runs through the property.

The nearest properties that may have salamander populations currently present, and where there have been historical sightings, are all directly inaccessible to the NLT site due to fragmentation. These known populations cannot naturally migrate, and the land cannot be linked in a practical means since it is many kilometers away to where the populations are seen on the Amphibian Atlas. Such fragmentation currently prevents salamanders from migrating (Shoop, 1968: Fccio, 2003). Though buffer zones could be created along the Cobourg Creek riparian zone, there is not enough habitat further up the creek to make that an advantageous property purchase in future (Figure 2). The disconnected and disjointed status of land in that area makes it difficult to choose a place for future connectivity planning. Regardless of which direction chosen the salamanders will need to cross many roads even if

more forest cover could be created in some areas. The area identified on the map as potentially worth connecting with is roughly 4km to the north-west of the Laurie Lawson property and has the largest current area of upland habitat. This area is recommended as having the best potential linkage habitat, if protective status and enhancement could be negotiated by the NLT with landowners.

It is recommended that the areas bordering Cobourg Creek be discussed with the current farm owners regarding mitigation of some of the effects of their runoff into the creek. This might be combined with discussing about the Atlantic Salmon restoration project and the potential for funding from both SARFIP and the Department of Fisheries and Oceans. Larger vegetation buffers would improve water quality for the fish and create connectivity for the salamanders on the Laurie Lawson property.

Longevity of Pools

Vernal pools are the sole breeding habitat used by spotted, blue spotted and Jefferson salamanders. The abundance of these pools on the property owned by the NLT is the primary reason for the discussion about moving salamanders to this property. Vernal pools do not alone provide enough habitat for breeding and developing salamanders but are still very important for success of some of the proposed species. The length of time which these pools remain is one of the principal focus points of our research. According to literature reviewed it is a primary concern for researchers in understanding salamander reproductive success. One of the experts interviewed said that salamander larval development in the vernal pools is directly related to the length of time the pools are present (Appendix 1). This is latitude dependant because of the seasonality and the associated atmospheric deposition. Adults will occupy the pools as soon as the snow has melted. Temperature affects the maturing larval

salamanders. The amount of time needed for salamanders utilizing vernal pools to develop will vary regionally (Appendix 1). In general, for Southern Ontario, developing salamanders will require the pools 55-70 days after the adults have left (Appendix 1). The literature shows that the length of hydroperiod was much more significantly related to success in development of larval salamanders than most other factors (Pechmann et al., 1989). The hydroperiod length was more significant to reproductive success than the number of reproducing females (Pechmann et al., 1989).

Methods

The site was initially scanned completely on foot by the 31st of May 2017, and vernal pools that were spotted had their spatial coordinates noted. Leaf litter and logs were also overturned to verify the lack of salamanders. A total of 9 trips were made. 2 trips were made in May to complete the inventory. 2 more trips were made on the 18th and 22nd of July to measure pool depths. Between the 14th and 29th of August, 4 more trips were made to measure vernal pool depths, pH of remaining vernal pools and surveying the forest. The final trip was made on the 5th of October 2017, to measure depths of remaining pools and removing any of the markers that remained.

Hydrology

Due to the irregular shapes of the pools, debris in the basin, inaccessible pool edges and a GPS resolution of 5 to 7m. The smallest of the vernal pools had a diameter of <1m while others were >10m. A lot of the pools had trees on their edges bending in either due to wet soil or to fill the canopy gap over the pool, creating varying shapes for surface area. During depth measurements, several logs were experienced in the basin creating varying

basin curvatures. Hence, the attempts at measuring canopy cover, surface area, or basin curvature were abandoned for modelling and, only the depths could be reliably measured.

The presence of two other organisms was monitored as indicators for site suitability; Green Frog (*Lithobates clamitans*) tadpoles and dragonfly (order *Odonata*) larvae. Green frogs share the same biological limitations and sensitivity common across amphibians, whereas dragon fly larvae are predatorial and their maturation indicates presence of benthic invertebrates that can act as food for the salamander juveniles. The pH for 21 of the pools was measured in the month of August. By this part of the season, the smaller pools with poor hydroperiods had dried up and, sufficient amount of deposition and release from the anaerobic decomposition in the basins had occurred. A handheld pH meter was used for these measurements. The device was calibrated prior to the start of the readings.

The terrestrial habitat was divided into a grid of 50x50m resolution, resulting in 155 possible sampling points. Using a random number generator (random.org, 1998), 20 sampling points were generated. Only 16 of these points were accessible due to the stream acting as a barrier. One of these was not sampled for terrestrial habitat because it was inside the largest vernal pool, pool 400, and the point could not be adjusted within any reasonable distance from the actual point without overlapping into the next point on the grid. Hence, there were a total of 15 sampling points to measure the terrestrial habitat. Randomness was chosen over a grid method to prevent any systematic error and to allow for the reasonable shifting of sampling points without break in any pattern, when encountering trails and vernal pools. At each site, $100m^2$ were sectioned off to count Coarse Woody Debris (CWD). The most representative spot in the site was chosen to measure canopy cover using a Densiometer. This spot was also chosen to collect a soil sample of the first 5 cm of the soil, which was stored in

air-tight capsules, since it would have a representable amount of interaction with the detritus. Figure 1 in the previous section, shows the sampling points used.

In total, 15 soil samples were collected. They were weighed, dried for 24 hours at 90 degrees C and reweighed. This provides us with the amount of water lost by the soil to evapotranspiration in the field. Then 2 mL of this dried soil was put in test tubes and subjected to de-ionised water at 0.250 mL increments. Soils were considered saturated when the next increment resulted in standing water for more than 15 minutes and then subsequently used to measure the pH. The purpose is to have a quantifiable upper limit of saturation for calculating soil moisture. The remaining soil from the samples was then weighed and subjected to a Loss on Ignition test for 6 hours at 450 degrees C, after which they were re-weighed. Due to the design, the lack of replicates makes it is unreliable to interpolate spatial spread of the soil, but this data is useful to set site parameters and witness a scale of variability on the site.

Using Thorthwaite's equation, we calculated the monthly potential evapotranspiration (PET). Climate data for the historic years from 2007 till September 2017 were obtained. A summary of the data is in appendix 2. Using the daily temperature data, annual heat indexes were calculated. A linear projection of the annual heat indexes was used to predict the heat index for 2017. Linear slopes between the monthly PET values were used to estimate daily values of PET. Appendix 2 tabulates the calculated PET and relevant variables and the linear projection used to predict for 2017.

We used the equation, **Equation 1=** $\boldsymbol{\Delta D = PPT - PET \pm G}$, to calculate observed PET where $\boldsymbol{\Delta D}$ is loss in water column, ***PPT*** is precipitation, ***PET*** is potential evapotranspiration and ***G*** is loss or gain from groundwater. We presumed ground water influence to be zero. Loss by infiltration is assumed to be an unchanging constant. Despite the change in height,

since the pools are not too deep, the effect of column height on change in infiltration rate is ignored. Hence, the climatic PET is expected to drive the PET trend and exhibit with respect to change in depth. Observed PET is also termed as observed loss' in this paper since it a measure of loss of height of the water column for any and all pools.

The observed water loss, calculated from equation 1, was then compared to see if it was compliant with Thorthwaite's PET. Thornthwaite's PET, was also used as the projected PET for the soil moisture tabulation. Since soil moisture cannot be negative and the upper limit is observed through the testing in lab, we can use precipitation data to see when the soils were last saturated. By comparing the amount of soil moisture lost between then using the projected PET, to the collection date, we can get a factor that describes in percentage, the compliance of the observed PET to the projected (Thornthwaite's) PET for each sample/sampling site.

Results

The preliminary search for pools and salamanders revealed 116 vernal pools, but no salamanders were observed on the property during any part of the study. The habitat showed signs of forest management and maintenance due to the school board's outdoor centre activities. We were informed that the site is undergoing removal of dog-strangling vine (*Cynanchum rossicum*) using chemical means, as well as, Buckthorn (*Rhamnus cathartica*) through physical removal and burning. During our research Garlic Mustard (*Alliaria petiolata*) was also encountered on the property. The site is rich with logs and showed signs of good moisture retention around the logs and leaf litter which can be seen over most of the property. We were informed that dog owners bring their dogs to the property for its trails, and this was witnessed on many occasions during our visits. A few small mammal burrows were

encountered throughout the property, especially in the north-eastern half of it. Garter snakes were observed on site.

Pool depths were noted, the depths being driven by precipitation. No springs were found. Vernal pools' identifying coordinates are listed in Appendix 3. Figure 3 compares precipitation, modelled PET and observed loss of water for the sampling periods, based off the average of all the pools with standing water. The observed loss of pool depth follows the trend of precipitation. This is consistent with Brooks (2014) observation that vernal pool hydrology was dictated by precipitation in central Massachusetts.

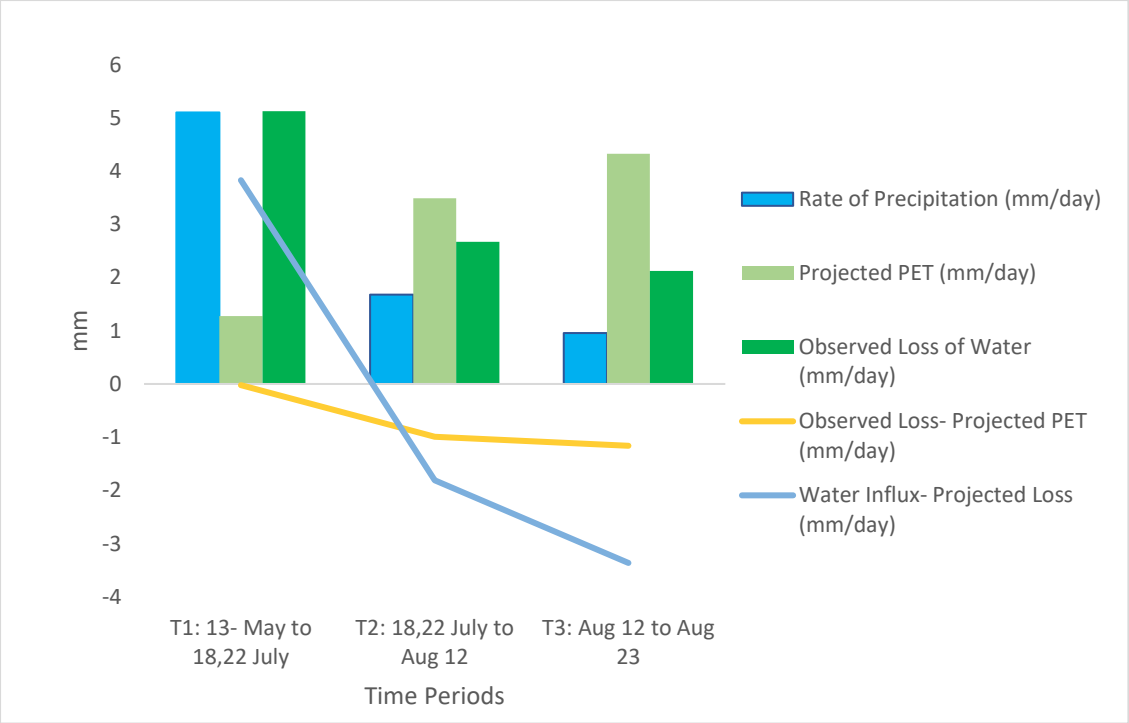

Figure 3: Observed precipitation, projected PET and observed loss of water from the vernal pools, for the three major observation periods, averaged across all vernal pools at the site. Number of pools measured for T1, T2 and T3 were 117, 86 and 54 respectively.

Historical weather data was compiled for the region from January 2006, until September 2017. A summary of the data is tabulated in Appendix 2. The precipitation received between April and August 2017, represents the period of this study. Table 1

tabulates this value for the last 11 years, with insufficient data being available for the site recorded for the year 2014.

Table 1: Annual precipitation summaries from 2007 to 2017.

Year	Sum of precipitation from April to September (mm)	rainfall per month (mm)
2007	127.8	18.26
2008	186	26.57
2009	254.5	36.36
2010	222.3	31.76
2011	283.4	40.49
2012	189.3	27.04
2013	247.3	35.33
2014	*Insufficient data*	*Insufficient data*
2015	150.3	21.47
2016	78.8	11.26
2017	539.3	77.04

The year of this study, 2017, had the highest rainfall, 539.3 mm, for the period of 10- years. This was almost twice the previous high value of 283.4 mm of rainfall. The hydroperiods of the vernal pools observed during the study were more exaggerated than they usually would be because of this extra rainfall. A concluding trip was made in October 5 and any pool with standing water remaining was identified as permanent for 2017. Historic weather trends (Figure 4), show that during the month of October the precipitation volume tends to increase again. Hence, it can be concluded that any pool that stays hydrated until October, will remain hydrated during winter.

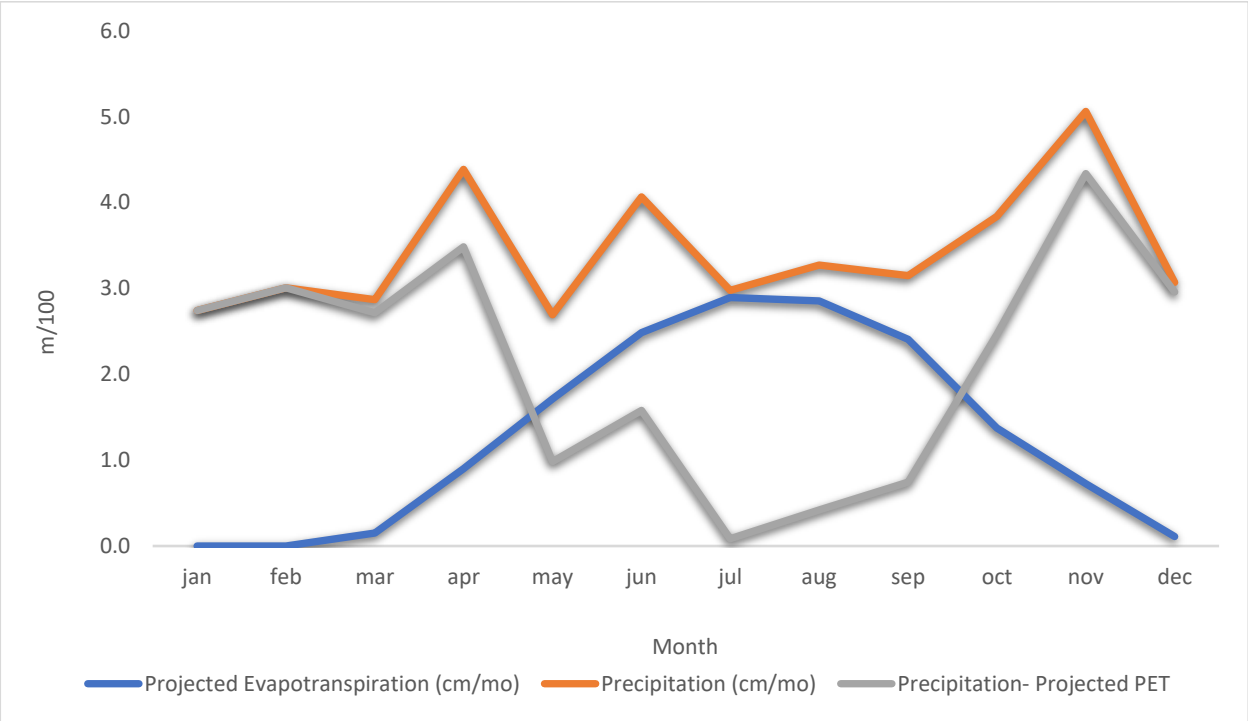

Figure 4: Plot of monthly climatic trends for projected PET and precipitation based on 9 years from the prior 10 years (2007- 2016) climatic period.

Table 2 summarizes the observed hydrology for the pools. Fourteen of the pools are marked as questionable. They either dried before the first measurement or were missed during measurement after the initial inventory. There were several pools that were deemed invalid because they either disappeared after our first survey and could not be measured again. Or other confounding reasons where the data was not valid. Pool 400 is the largest pool and was initially suspected to be a permanent pool but on October 5, had receded to a small patch with only 5 cm of standing water. Pool 52 retained the deepest standing water at 64.5cm on the 29th of October. The pool numbers in Table 2 that are highlighted in green, mark vernal pools that were also witnessed to house adult frogs and larvae.

Figure 5 maps all the sampling points as well as the vernal pools with respect to the observed hydroperiods. Corresponding coordinates for all sampling points are in Appendix 3.

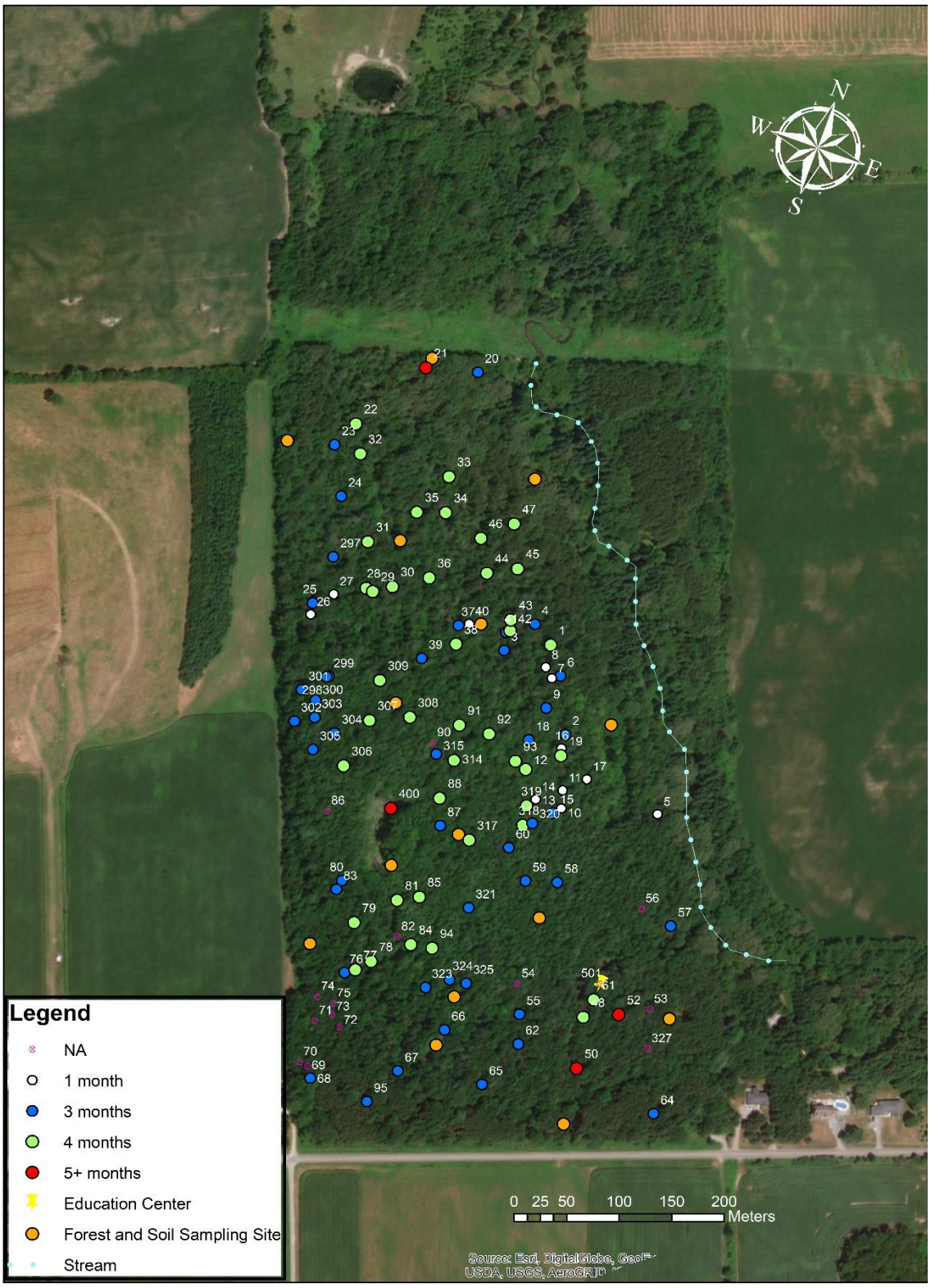

Figure 5: A map of all the vernal pools and forest sampling points. The vernal pools are labelled and categorized by hydroperiods. Refer to Appendix 3 for coordinates of points.

TABLE 2 : Observed Hydroperiods for the vernal pools	
Hydroperiod	Pools (by number)
1 (-May 27)	5, 7, 8, 10, 11, 13, 14, 16, 17, 26, 27, 40
3 (-July 22)	2, 3, 4, 6, 9, 15, 18, 20, 23, 24, 25, 37, 39, 41, 55, 57, 58, 59, 60, 62, 64, 65, 66, 67, 68, 76, 80, 83, 87, 95, 297, 298, 299, 300, 301, 302, 303, 304, 305, 315, 320, 321, 323, 324, 325.
4 (-August 29	1, 12, 19, 22, 28, 29, 30, 31, 32, 33, 34, 35, 36, 38, 42, 43, 44, 45, 46, 47, 48, 61,77, 78, 79, 81, 84, 85, 88, 91, 92, 93, 94, 306, 307, 308, 309, 314, 317, 318, 319.
6 (- October 5)	400, 21, 50, 52.
Invalid	53, 54, 56, 69, 70, 71, 72, 73, 74, 75, 82, 86, 90, 327.

The indicator species of dragonflies (sub- order Anisoptera) and frogs (primarily: *Rana clamitans*) showed successful metamorphosis. Other frogs were also seen, but their breeding success was uncertain. The pools with standing water on August 23 were measured for pH, and showed a range of 7 to 7.7, with an average of 7.3.

The forest attributes were parameterized through 15 randomly generated points. Appendix 3 holds all the values for the sampling points along with a labelled map. In general, we found that the canopy cover ranged from roughly 99% to 88.5%, with an average of approximately 95%. However, the canopy showed gaps from trees falling or being felled. A survey of the woody debris showed that on average, there were 4 logs with a category 3 or 4 level for every 100m^2. Levels 3 and 4 are stages where the woody debris has decomposed to the point of breaking the whole log into smaller splinters or fibers, respectively. At these stages of decomposition, the woody debris is able to retain moisture and air, in turn provide shelter to salamanders. There was only one site which had neither a category 3 nor category 4 woody debris. However, there were 3 category 2 woody debris present there, showing future presence of category 3 and 4 from those pieces. These values were counts of logs from individual woody entities and not counts of pieces of the same (former) tree or the branches. The soil

samples showed a pH range of 6.09 to 7.60, with an average of 6.85. The loss on ignition test showed organic content ranging from 5.5% to 29.6%, with an average organic content of 13.6%. The average amount of water 1mL of soil could hold, at a resolution of 0.25mL, was 0.75mL. The soil moisture balance that used soil moisture from the soil samples to correct the modelled potential evapotranspiration, showed that between April and August of 2017, there was no dry days i.e. a day with 0 soil moisture. The soil samples are indicative of only the top 5cm of the soil and provide a range for this site. Interpretive data from all the sampling points is listed in Table 3.

Table3: Data from terrestrial habitat evaluation sampling points

	Coordinates (UTM83)		Coarse Woody Debris Count (/100m^2)				Canopy	Soil Sample Analysis		
#	Longitude (m)	Latitude (m)	C1	C2	C3	C4	Cover (%)	% Organic (g)	Saturation (mL)	pH
1	723091.1021	4874193.971	0	3	3	1	88.54	12.04	0.875	6.56
2	722950.6941	4874224.651	1	2	1	3	94.79	9.46	0.75	6.76
3	722804.0592	4874276.461	2	3	4	1	91.67	12.47	0.625	6.58
4	722950.6941	4874274.651	2	2	4	4	94.79	9.61	0.875	7.22
5	723150.6941	4874324.651	1	3	5	5	92.71	7.90	0.625	6.7
6	723000.6941	4874374.651	2	0	3	2	93.75	5.47	0.625	6.41
7	722850.6941	4874374.651	0	2	1	1	93.75	18.52	0.625	6.56
8	722900.6941	4874424.651	1	2	3	2	98.96	11.25	0.875	7.37
9	722800.6941	4874524.651	1	0	3	3	97.92	23.26	0.75	6.09
10	722850.6941	4874624.651	2	2	2	2	95.83	13.43	0.875	7.24
11	722750.6941	4874674.651	1	0	0	2	95.83	10.66	0.625	6.67
12	722616.3664	4874729.447	0	5	3	3	95.83	8.42	0.75	7.45
13	722718.3971	4874851.882	0	3	0	0	95.83	16.92	0.625	7.6
14	722850.6941	4874774.651	2	4	1	0	95.83	14.84	0.75	6.7
15	723000.6941	4874574.651	0	1	1	2	97.92	29.61	0.875	6.84

Note: Coarse woody debris category C1 is completely intact logs, C2 is logs without branches, C3 is logs whose cork is splintering and C4 is logs whose cork has turned fiberous.

Habitat Suitability

The site could be considered to have sufficient habitat to host a salamander/newt population. Though the issue of connectivity could likely become a hindrance, and potential genetic island causing. This section will interpret the literature review, expert review and results from the field observations in accordance to potential application.

Species and Populations

Red-backed salamanders have potential to be territorial especially for foraging lands and can show aggression to other members of their species. There is some evidence that territorial behaviour is more prominent as an issue for members of the same species, especially with Spotted Salamanders (Walls, 1990). Hence, the landscape area will limit the population size of the species, especially the territorial species which will limit themselves more aggressively.

There is also potential for the adult Central Newt but as mentioned, it will eat other salamander larvae and egg masses. Hence, it would be detrimental to introduce the newt on the NLT site along with the salamanders. While they will be very well suited for the site, especially since they can also occupy slow moving streams, they will prevent the success of the pool dependant salamanders. It is not recommended to introduce them along with the Spotted or Blue-spotted salamanders; only in isolation or with the Red-backed salamanders. It must be noted however, that despite their colonising nature and ability to live in streams, none were found to have colonized the site, despite the recent colonisation by frogs. There may be other factors related to why the Central Newt has not already colonized the area, or they may already exist in the stream which was not surveyed in detail in this study.

Modes of Reintroduction and establishment

Mike Oldham at the Natural Heritage and Information Centre, in his conversation with us, mentioned that the MNRF will not assist in moving species or individuals. The NLT would then need to find assistance for the reintroduction to the site through private means. One option would be to contact builders in nearby locations who might be clearing wetlands or pools that have the targeted salamander species and move their populations with assistance from the builder(s).

The other option, as mentioned in the introduction and in all the interviews, was the reestablishment of landscape connectivity to promote more natural migration of salamander populations to the site. This is as noted in the introduction, a very large challenge due to the landscape being highly fragmented. Even if the buffer zone along the creek is extended, it would have to be sufficient for the salamanders to migrate over to the site. The salamanders are known to migrate small distances in a range of around 200m while the creek's buffer zone links with habitats over 1000m away. Because of this the buffer zone extension would require considerable development of habitat along the route to enable migrations to the NLT site. It must also be noted that the Ontario Reptile and Amphibian Atlas has limited recording of salamanders from this area because few surveys have been undertaken, thus it is likely there is habitat with populations along the stream that have not been documented. Surveying of the creek length is recommended in-order to determine what habitat is required to promote the natural migration of salamanders to the site.

Monitoring

Monitoring of the salamander population and habitats would be needed to verify success of the salamander populations established, irrespective of how they reach the site.

The parameters mentioned in the prior sections for the hydroperiods, water quality, soil quality and forest cover are essential to support the salamander populations. Thus, they should be monitored. Another indicator would be surrogate species. Frogs like the green frog, are a good indicator of water quality. Figure 6 is a graph that shows that frogs and tadpoles

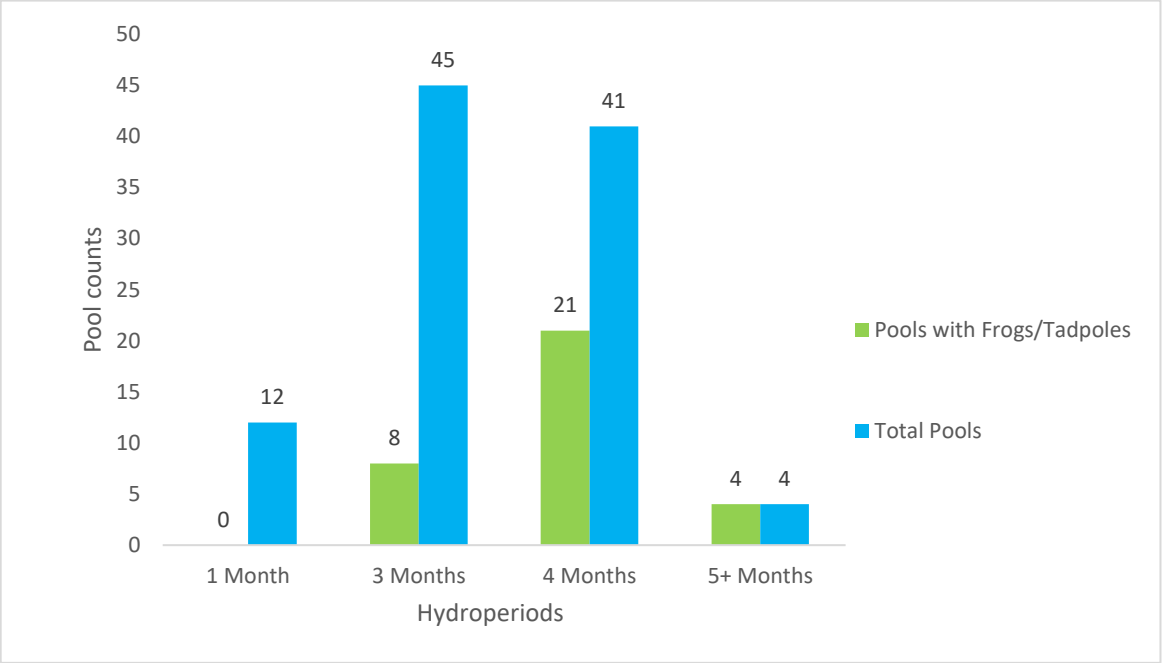

Figure 6: A graph of pools observed with frogs and/or tadpoles, for each category of observed pool hydroperiod. A larger fraction of the pools with larger hydroperiods were observed to house frogs/tadpoles.

were likely to exist in pools with longer hydroperiods.

The success of frogs in a vernal pool indicates that the pool can most likely host salamander larvae too. A larger portion of the pools with longer hydroperiods are occupied by these frogs. This can be considered an indicator of the potential success of any salamander population with breeding pool fidelity. Pools with longer hydroperiods will have more success and hence have a higher preference later in the season, causing the populations centred around them. Populations of vernal pool breeding salamanders will be most abundant in the pools with the longest hydroperiods. To be very successful in sustaining populations the distance around the pool for foraging habitat needs to also be considered. Pools with a 200m perimeter of upland habitat for the Spotted and Blue-spotted Salamanders could be

targeted for ongoing monitoring; based on a study of Spotted Salamanders, though the adults occupied only approximately 113m on average (Faccio, 2003: Ryan & Calhoun, 2014). For any Red-backed Salamanders, the forest canopy cover and forest floor cover density will dictate prevalence. A shaded and well-littered forest floor provides shelter during the day as well as food from the invertebrates and insects in the litter. Adults of the Spotted and Blue-spotted salamanders will also have similar dietary and cover requirements, but they may also use small mammal burrows for refuge.

Figure 5 shows the distribution of pools and their hydroperiods. In the year the observations were made, 2017, the precipitation was abnormally more abundant during the summer. This implies that during a year with normal precipitation, the vernal pool hydroperiods will be shorter than observed in 2017. The vernal pools with four or more months of hydroperiod are estimated to be ideal for salamanders in years of normal precipitation. Figure 5 displays regions in the site where salamanders are more likely to reside. The southern edge, for example, has a cluster of vernal pools with relatively smaller hydroperiods and perhaps less potential in low precipitation years. The terrestrial salamanders without an aquatic phase, will spread according to habitat and food. Monitoring the populations will allow proper measure of the spatial spread.

To measure size and success of the salamander populations, monitoring of live adults using coverboards can be used to quantify and the population. The early spring time with above freezing temperatures, is ideal for this. Untreated coverboards are placed in a sampling grid. The boards usually remain unoccupied for several years before developing conditions underneath that would be suitable for salamanders. The boards will have to soften up from decomposition and moisture to the extent that they become favoured as resting habitat by

salamanders. The coverboards should not be more than 12x12 inches as large boards lead to dry soils. Irrespective of final size, the arrangement and monitoring management of the boards should be consistent. Leaf litter should be cleared from the site and the coverboards placed directly in contact with the ground. A weekly schedule can be used to collect the salamanders under the boards. They must be collected using gloves and held from the centre or 'waist'. They must be stored in slightly moist containers, identified and measured and returned to the same coverboard as soon as possible. Care must be taken to not hurt the salamanders when placing the coverboards back on them. (Fernandes R, 2002) An annual study until the population size stabilizes, followed by a 3- or 5-year monitoring cycle can allow checking spread, success and fitness of the salamander population.

Management

Post-introduction, the maintenance of the NLT site will have to incorporate measures to accommodate the salamanders. The forest habitat will have to be maintained to ensure that canopy cover does not fall below the critical thresholds mentioned in the introduction. The current practice of logging dead trees and keeping the tree trunks on site has provided the site with coarse woody debris that is beneficial to adult salamanders and should be continued. However, care with forestry practices will have to be taken to ensure that runoff from adjacent lands and dog-strangling control measures do not contain of chemicals that can harm the salamander population. Also, while the site's vernal pools could currently support a salamander population, the future transformation of the pools must be monitored. The organic debris that accumulates in the pools will alter the basin and in turn, alter the hydrology of the pools. The vernal pool hydrology might be required to be observed for a longer period to

understand the details of the trend. Depending on the trend of the site, a future need to intervene and alter the vernal pools might arise.

The site is self-contained with respects to the habitat and breeding sites because of this, it is possible that the migrations would not involve the salamanders travelling outside the property. However, the possibility exists, and future care might be required to minimize fatalities through awareness and signage on the road or building physical barriers. The stream within, and clearings outside the property, will act as natural barriers during the non-migratory phase, though some individuals might explore the surroundings. Overall, the site is expected to mostly contain the populations within itself. The fragmentation of the landscape with respect to naturalized areas, in this regard, is beneficial in terms of containment. This could also result in some detrimental genetic effects. For this reason, it is our biggest recommendation to pursue options and possibilities for connectivity.

Should the newt be established, the colonising nature of the Central newt must be considered. The stream/creek can act as a passage for the newts to colonise other landscapes and vernal pools. Their predatory nature might affect the frog populations as well. Also, the stream quality will also need to be considered since agricultural runoff from surrounding land parcels might harm the newts.

Several opportunities for research and education arise from the introduction of the salamanders. Population dynamics and spatial spread will itself yield a lot of interesting observations as the population establishes itself. The food chains in the forest will also alter since the salamanders are both prey and predators themselves. The vernal pools themselves can be studied. The water balance, benthic invertebrate composition, seed bank transformation over the last 60 years, gaseous intake and release and the long-term transformation of the vernal pool hydrology, are some of the topics that the site can be used

to research for. Apart from research, salamanders themselves can act as an educational tool to involve and teach students at the Laurie Lawson education centre. Introduction of salamanders to the property, with great care and research, could be an amazing project for the NLT.

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Appendix 1 : Expert Interview Text

Madison Wikston – Masters Student at Trent – TA for Herpetology class

S- In your research did you find any specific characteristic in or around a pool that guaranteed salamander presence?

M- mainly longevity of the pool, if the water was deep enough and remained for long enough during the season for the larval salamanders to develop and emerge. As well, the presence of leaves in the bottom of the pond and not too dirty of water.

S- Would you say there is a landscape component that would make salamanders proliferate in an area?

M- Canopy cover will affect how long the pools remain or evaporate which will affect success rates, as well as in the area being studied specifically the large amounts of cow farms and other agricultural lands surrounding could negatively affect mortality rates for salamanders.

S- To your knowledge what salamander species would be best suited to this area of Ontario?

M- Jefferson's, spotted, and blue spotted. Red backs are more widespread and easily do well most places but do not require vernal pools.

S- Based on the research question would you have any recommendations for our study, that perhaps were things you found helpful in your pond studies previously?

M- consider taking extra measurements in the field, such as including a visual encounter survey but simple to see if any salamanders are already present and if not perhaps other amphibians or wildlife. Also, the area that is owned by the Northumberland Land Trust in that area is quite isolated and salamanders need space to migrate even small amounts for foraging etc. looking for areas to gain continuity with surrounding naturalized areas would be helpful. Introducing species may not do well as the area is very high in traffic.

S- With the NLT's desire for salamanders to be moved to this area, do you think this would be approved by the MNR for assistance?

M- I'm not sure that the MNR will assist with that, I am not sure of their policies regarding moving of amphibians. But I don't think it is a usual occurrence.

Patrick Moldowen – PhD candidate at University of Toronto – Studying populations of salamanders in Algonquin Park

S- How long do developing salamanders occupy vernal pools in Ontario?

P- Length of time used for developing salamanders is largely latitude dependant. Temperature greatly affects the maturing larval salamanders and thus the window of time needed for salamanders varies regionally. If the pools dry up too early the salamanders will die, and in general the pool availability will not last longer than the needed time. In general for Southern Ontario developing salamanders will require the pools 55-70 days after the Adults have left.

S- How long do the Adults require the pools and when do they move to the pools for breeding?

P- The adult salamanders will really occupy the pools as soon as the snow has melted. They will likely occupy the space for 3-4 weeks which can often overlap with the larvae beginning to hatch.

S- Do you have any other recommendation or anything major to consider for suitability of habitat for salamanders?

P- Connectivity would be the biggest issue other than having basic appropriate conditions. If the space is available and the adults can move into the area, then they generally will occupy the space. So making sure movement to upland habitat is possible will be key.

Mike Oldham – Natural Heritage and Information Centre

S- I was referred to you through Tom Whillans because of a community based research course that another student and myself are conducting. We have been requested by the NLT to survey and asses one of their properties for suitability of habitat for salamanders, so that they might request assistance from the MNR to eventually have some moved to their property to help with population decline. Are you aware of any specific quality parameters to look for that would be best for salamanders?

M- The MNR does not move animals almost ever. So, I don’t know how else I might help.

S- Is there anything then we might consider continuing with the project for determining suitability?

M- You could look at habitat continuity and connectivity to try to help facilitate the salamanders naturally migrating to that property. There are a lot of different programs and resources for landowners to get tax breaks etc. to let parts of their property naturalize or to replant trees in an area. I would look at a map and survey the areas surrounding to see why salamanders aren’t already there.

Tina Fridgen – Professor at Trent – research in amphibians, teaches Herpetology

S- Is there any recommendations you might have regarding suitability of habitat for salamanders that would be most beneficial to repopulating an area?

T – My research is mainly to do with frogs, but I would recommend looking at the surrounding areas. Get a map and look at google earth or other satellite images to see options for connectivity. You may also want to consider how runoff from the surrounding farms will negatively affect the salamanders. Also, just to note the MNR does not move animals, almost ever. You'd have to create a pathway for them to migrate on their own.

S- Is there any qualities that you know of specifically for vernal pools that would be better suited for salamanders, should they migrate on their own?

T- I would check for favourable pH and general water quality, if you're getting runoff from the surrounding farms this may be why there currently aren't salamanders already.

*Note the remainder of time in the interview was spent looking at satellite images where Tina recommended areas to look for partnering with farmers or the NLT to buy portions of property to naturalize to encourage more migration pathways.

Appendix 2: Summary of data used for climatic evaluation of site

TABLE 2A: Annual summary of Precipitation and PET between April and August for 2007 to 2017			
YEAR	Sum of PPT, April to August (mm)	Sum of PPT, June to August (mm)	average monthly precipitation (April to August) (mm/month)
2007	127.8	64.8	18.3
2008	186.0	152.1	26.6
2009	254.5	106.3	36.4
2010	222.3	165.3	31.8
2011	283.4	156.1	40.5
2012	189.3	123.1	27.0
2013	247.3	136.0	35.3
2014			
2015	150.3	89.4	21.5
2016	78.8	38.6	11.3
2017	**539.3**	**312.4**	77.0

TABLE 2B: Heat indexes to use in Thornthwaite's PET equation.	
Year	Heat Index
2007	17.13
2008	17.34
2009	17.38
2010	21.32
2011	19.23
2012	
2013	
2014	
2015	18.54
2016	22.63
2017	21.72

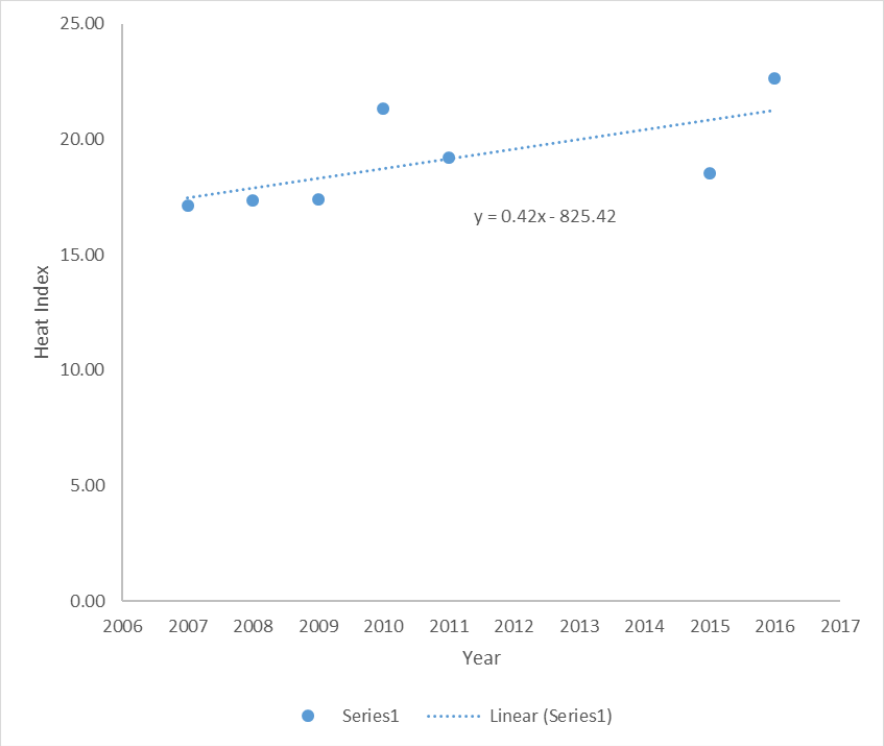

Figure 2A: Linear plot of heat annual heat indexes used to estimate the heat index for 2017. An increasing trend is seen for heat indexes over this 10 year period.

Appendix 3

Table 3A: List of all the sampling points.			
ID	Latitude (WSG84)	Longitude (WSG84)	Type
1	43.99092	-78.22	Pool
2	43.99022	-78.2195	Pool
3	43.99075	-78.2205	Pool
4	43.99105	-78.2203	Pool
5	43.9898	-78.2182	Pool
6	43.99069	-78.2198	Pool
7	43.99065	-78.2199	Pool
8	43.99072	-78.22	Pool
9	43.99038	-78.2198	Pool
10	43.98959	-78.2193	Pool
11	43.98974	-78.2193	Pool
12	43.98982	-78.2198	Pool
13	43.9896	-78.2196	Pool
14	43.9896	-78.2196	Pool
15	43.98953	-78.2194	Pool
16	43.99009	-78.2195	Pool
17	43.9899	-78.2191	Pool
18	43.99007	-78.2199	Pool
19	43.99003	-78.2195	Pool
20	43.993	-78.2219	Pool
21	43.9929	-78.2225	Pool
22	43.99225	-78.223	Pool
23	43.99202	-78.2232	Pool
24	43.9916	-78.2229	Pool
25	43.99064	-78.2229	Pool
26	43.99054	-78.2228	Pool
27	43.99077	-78.2227	Pool
28	43.9909	-78.2223	Pool
29	43.99089	-78.2222	Pool
30	43.99099	-78.222	Pool
31	43.9913	-78.2225	Pool
32	43.99201	-78.2229	Pool
33	43.99205	-78.2218	Pool
34	43.99175	-78.2217	Pool
35	43.99167	-78.222	Pool
36	43.99116	-78.2216	Pool
37	43.99084	-78.2211	Pool
38	43.99068	-78.2211	Pool
39	43.99047	-78.2214	Pool
40	43.99097	-78.2247	Pool
41	43.99091	-78.2206	Pool

42	43.99093	-78.2205	Pool
43	43.99102	-78.2206	Pool
44	43.99135	-78.221	Pool
45	43.99147	-78.2207	Pool
46	43.99163	-78.2212	Pool
47	43.99183	-78.2209	Pool
48	43.98791	-78.2182	Pool
50	43.98746	-78.2181	Pool
52	43.98802	-78.2179	Pool
53	43.98815	-78.2175	Pool
54	43.98801	-78.2191	Pool
55	43.98776	-78.219	Pool
56	43.98896	-78.218	Pool
57	43.9889	-78.2176	Pool
58	43.98896	-78.219	Pool
59	43.98889	-78.2194	Pool
60	43.98912	-78.2197	Pool
61	43.98808	-78.2182	Pool
62	43.98751	-78.2189	Pool
64	43.98729	-78.2171	Pool
65	43.98708	-78.2191	Pool
66	43.98743	-78.2198	Pool
67	43.98697	-78.2201	Pool
68	43.98668	-78.2211	Pool
69	43.98677	-78.2212	Pool
70	43.98678	-78.2213	Pool
71	43.98716	-78.2213	Pool
72	43.98718	-78.221	Pool
73	43.98726	-78.2211	Pool
74	43.98737	-78.2213	Pool
75	43.98736	-78.2211	Pool
76	43.98764	-78.2211	Pool
77	43.9877	-78.221	Pool
78	43.98781	-78.2208	Pool
79	43.98809	-78.2212	Pool
80	43.9884	-78.2215	Pool
81	43.98839	-78.2208	Pool
82	43.98808	-78.2206	Pool
83	43.98832	-78.2215	Pool
84	43.98805	-78.2205	Pool
85	43.98847	-78.2205	Pool
86	43.98894	-78.2219	Pool
87	43.98912	-78.2206	Pool
88	43.98935	-78.2207	Pool
90	43.98979	-78.221	Pool
91	43.99001	-78.2207	Pool

92	43.99002	-78.2204	Pool
93	43.98986	-78.22	Pool
94	43.98808	-78.2202	Pool
95	43.98663	-78.2204	Pool
501	43.98823	-78.2182	Camp
297	43.99108	-78.2228	Pool
298	43.99007	-78.2224	Pool
299	43.99006	-78.2224	Pool
300	43.98984	-78.2225	Pool
301	43.98989	-78.2227	Pool
302	43.98961	-78.2226	Pool
303	43.98969	-78.2224	Pool
304	43.98961	-78.2221	Pool
305	43.98942	-78.2223	Pool
306	43.98936	-78.2219	Pool
307	43.98981	-78.2218	Pool
308	43.98994	-78.2213	Pool
309	43.99017	-78.2218	Pool
314	43.9897	-78.2207	Pool
315	43.98971	-78.2209	Pool
317	43.98908	-78.2202	Pool
318	43.98935	-78.2196	Pool
319	43.98952	-78.2197	Pool
320	43.98939	-78.2195	Pool
321	43.98852	-78.22	Pool
323	43.98774	-78.2201	Pool
324	43.98787	-78.2199	Pool
325	43.98788	-78.2197	Pool
326	43.98803	-78.2183	Pool
327	43.98781	-78.2174	Pool
11	43.98697	-78.2181	Soil
15	43.98728	-78.2198	Soil
23	43.98779	-78.2216	Soil
26	43.98773	-78.2198	Soil
41	43.98812	-78.2173	Soil
48	43.98862	-78.2191	Soil
45	43.98866	-78.221	Soil
57	43.9891	-78.2203	Soil
76	43.99003	-78.2215	Soil
98	43.99091	-78.2209	Soil
107	43.99139	-78.2221	Soil
115	43.99193	-78.2238	Soil
145	43.993	-78.2224	Soil
130	43.99226	-78.2208	Soil
91	43.99042	-78.219	Soil
400	43.9892	-78.2212	Pool

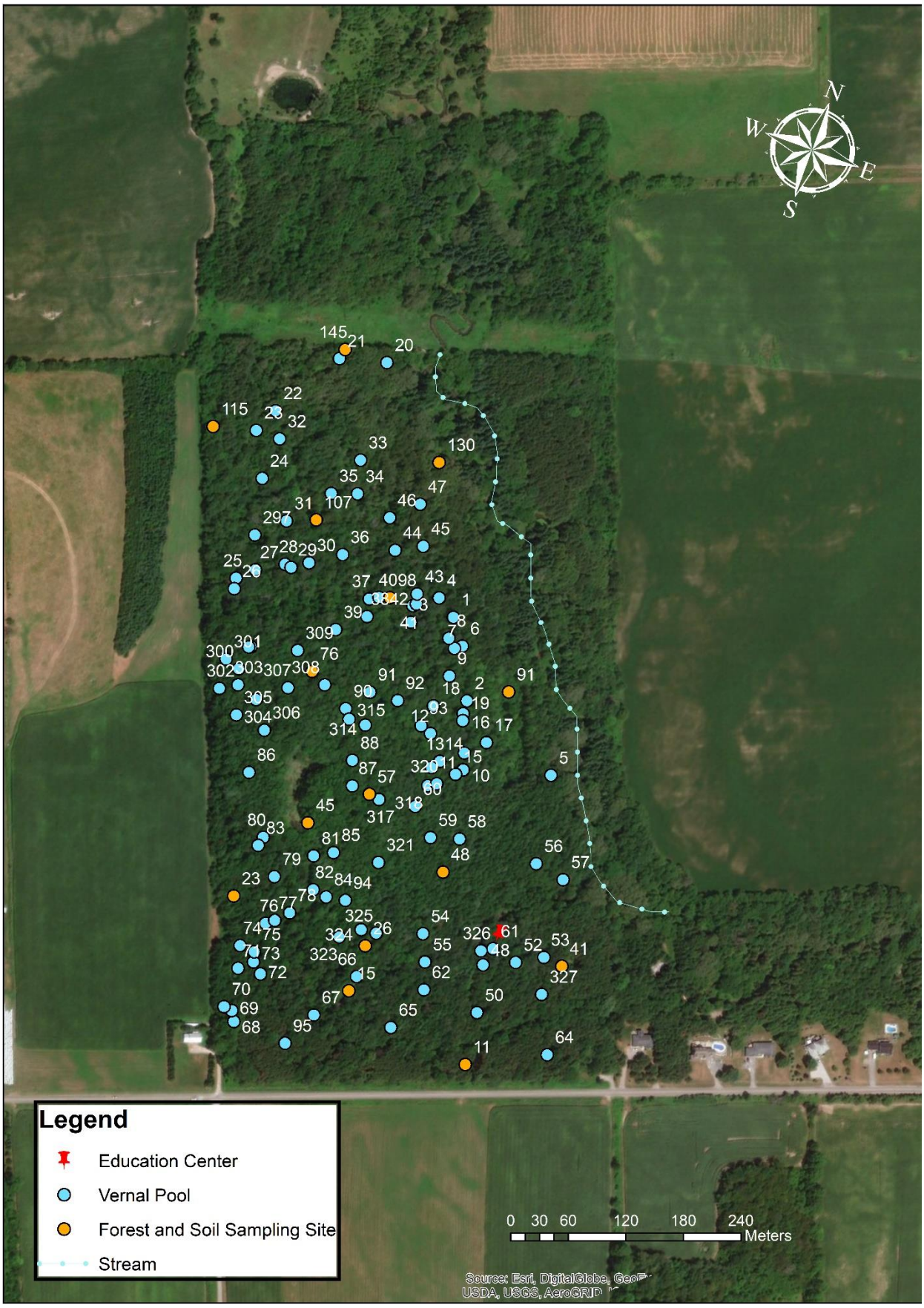

Figure 3A: A map with all the sampling points labelled. The 'Education Centre' does not have number labelled to it and exists solely as a marker